Ali Daher

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Qualifications

University of Oxford: Institute of Biomedical Engineering Oxford, UK
Doctor of Philosophy in Engineering Science: Oxford Institute of Biomedical Engineering *October 2020 - Present*

Massachusetts Institute of Technology (MIT) Cambridge, US
Mechanical engineering with a concentration in biological engineering. Concentration in literature (5.0/5.0) *June 2015- December 2019*

University of Cambridge Cambridge, UK
MIT Horizons Scholar: enrolled in Pembroke-King's Summer Program (First Class, 4.0/4.0) *July-August 2018*

Awards

- **Rhodes Scholar**: Selected as one of the two Rhodes scholars from Syria, Jordan, Lebanon and Palestine constituency for 2020.
- **MIT SUPERUROP program**: Selected as a scholar for SuperUROP, a competitive yearlong advanced research program. Awarded best undergraduate poster presentation for research on Glioblastoma multiforme (GBM) brain tumor growth modeling. Awarded certificate of advanced undergraduate research in biological engineering.
- **Finite Element Methods for Mechanical Engineers Competition (MIT)**: Awarded first place in class competition for the computational optimization of the tallest lighthouse structure that supports its own weight without mechanical self-buckling due to compressive stress.
- **Quantitative Systems Physiology (Harvard-MIT Division of Health Sciences and Technology)**: Awarded first place in class competition for best performance in the patient case study challenges.
- **MIT Burchard Scholarship**: The scholarship program brings together distinguished faculty members and 35 promising MIT students (nominated by their faculty), who have demonstrated excellence in the humanities, arts, and social sciences, for discussions and seminars.
- **MIT Abdel Latif Jameel Toyota Scholar**: Offered to support outstanding students and scientists from the Arab World admitted to MIT.
- **International Genetically Engineered Machine (iGEM) Team Competition**: Competed in the iGEM synthetic biology research competition with MIT's team. Awarded the iGEM Conference bronze medal for investigating the control of alternative splicing mechanism of mRNA.

Publications and Conferences

- 'A Dynamic Multiscale Model of Cerebral Blood Flow and Autoregulation in the Microvasculature.' Journal of Applied Mathematical Modelling (2023).
- 'A Network Model of Dynamic Cerebral Autoregulation.' Journal of Microvascular Research (2023).
- 'A Network Model of Dynamic Cerebral Autoregulation.' World Congress in Biomechanics (2022).
- 'A Dynamic Model of Autoregulation in the Cerebral Microvasculature.' International Conference on Computational and Mathematical Biomedical Engineering (2022).
- 'Mathematical Models of the Cerebral Microcirculation in Health and Pathophysiology.' International Conference on Computational and Mathematical Biomedical Engineering (2022).
- 'Bayesian Learning Machines for Glioblastoma Multiforme Brain Tumor Evolution.' SIAM Conference on Uncertainty Quantification (2022).
- 'MIT-Skywalker: Evaluating Comfort of Bicycle/Saddle Seat.' Proceedings of International Conference on Rehabilitation Robotics (2017).
- 'Detoxification Assessment of Inorganic Mercury by Bioluminescence of Vibrio fischeri.' The Journal of Environmental Toxicology (2017).

Teaching Experience

University of Oxford Engineering Science Department Oxford, UK
Tutor for Biofluids *January 2023 - Present*

- Conducted tutorials and review sessions for 18 students in groups of 3-4, marked their work, provided feedback on their work to the students and to the colleges/department, and helped with any individual areas of weaknesses.

MIT Department of Mechanical Engineering Cambridge, US
Undergraduate Teacher Assistant for Dynamics and Control I *February - May 2018*

- Graded problem sets and quizzes and provided feedback on a weekly basis for 75 students taking Dynamics and Control I.

Instatoot and University Tutor Tutoring Services Amman, Jordan
Tutor *December 2018 - August 2020*

- Tutored IB high school students and first year engineering and medical school students at Jordanian universities.

Work and Research Experience

MIT Multidisciplinary Simulation, Estimation and Assimilation Lab Cambridge, US
MIT SUPERUROP Research Scholar *August 2018 - February 2020*

- Updated the partial differential equation-based mathematical model for modeling Glioblastoma multiforme (GBM) tumor growth in the brain to incorporate into the MATLAB simulation the effects of a chemotherapy treatment plan.
- Adapted a novel stochastic modeling method, previously used by the lab for fluid systems, for scholastic modelling of the tumor evolution, thereby creating an accurate, reduced order scheme that captures uncertainty evolution at a significantly less computational cost than conventional ensemble forecasting methods.

Harvard Medical School - Brigham and Women's Hospital Divison of Genetics Boston, US
Project Leader - Researcher *June - September 2019*

- Led a project to develop a novel approach for better detection of pathogenic genetic mutations via deriving information from the 3D superimposition and structural alignment of the protein with its homolog and isoform proteins.
- Created a computational pipeline that extracts information from gene and protein databases and structurally aligns ortholog structures, and assesses the pathogenicity of a particular gene variant by feeding data into a trained classifier.

Shmeisani Hospital Amman, Jordan
Intern and Mentee *May - July 2018*

- Performed laboratory tests on patients' clinical specimen in the departments of microbiology, hematology and chemistry.

Synthetic Biology Center at MIT Cambridge, US
Undergraduate Research Assistant/ Member of MIT's International Genetically Engineered Machine (iGEM) Team *June - November 2017*

- Competed in the iGEM research competition, where each team applies synthetic biology to build a genetically engineered system from standard biological parts to tackle a world problem. Presented results in the iGEM Giant Jamboree Conference.
- Investigated the control of alternative splicing mechanism of mRNA using dCas13a and MS2 RNA Binding Proteins, the goal of which resulted in the building and testing of a collection of about 200 plasmids.

MIT Department of Mechanical Engineering Cambridge, US
Research Assistant at The Newman Laboratory for Biomechanics and Human Rehabilitation *September - December 2016*

- Designed and created different bicycle saddles that were installed on the MIT-Skywalker robot for the rehabilitation of the gait and balance of patients after a neurological surgery.
- Evaluated the different designs by analyzing the pressure data from the saddle and the responses of the human participants to determine the most suitable design for the patients.

MIT Energy Initiative Cambridge, US
Research Assistant at the MIT TATA Center *September 2016 - February 2017*

- Investigated the optimal reactor conditions for thermochemical treatment of biomass waste.
- Ran Thermal Gravimetric Analysis (TGA) experiments on different biomass fuels and analyzed their composition to determine the most energy-efficient, cost effective fuel to be used in rural areas in Kenya.

University of Hassan II Biotechnology Lab Casablanca, Morocco
Visiting Researcher as Part of MIT International Science and Technology Initiative *June - August 2016*

- Investigated the treatment of mercury-contaminated Moroccan waters using free and immobilized bacteria.
- Determined the most mercury resistant bacterial strain and the optimal conditions for the purifying performance.

Other activities

Rhodes alumni book club discussion facilitator, Rhodes warden's book club, Exeter college boat club, Oxford basketball varsity team, MIT Men's heavyweight crew team, MIT basketball intramurals, Palestine@MIT, Delta Psi fraternity, Jordan Orthodox basketball club, MIT triathlon club, Jordan national basketball team, MIT applicant interviewer, SJLP Rhodes outreach team, MIT Arab student organization, MIT international science and technology initiative, MIT international student organization, Caritas Jordan.